

Muskaan Irfan

Journal Publications (1 Published, 2 Submitted)

Silver Medalist

MS Data Science (3.68/4)

BS Maths (3.2/4)

MS/PhD Candidate 2026

https://www.linkedin.com/in/muskaan03/

muskaanirfn@gmail.com

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Cover Letter

Ms. Muskaan Irfan
BS Maths, MS Data Science
muskaanirfn@gmail.com

Subject: Looking for MS/PhD, MS DS (Silver Medalist), BS Maths, 1 Publication

Dear Sir/Madam

Hope you are doing well. My name is Muskaan, and I am writing to express my interest in MS/PhD positions with the following profile:

Education

- MS Data Science (Silver Medalist, CGPA 3.68/4) from Air University, Pakistan
- BS Mathematics (3.2/4) from Quaid-i-Azam University, Pakistan

Research Experience

- 1 paper published (International Journal of Applied Technology in Medical Sciences, Impact Factor: 3.5)
- 2 papers submitted
- Work in progress on 3 more papers

Other Notable Achievements

- Merit scholarship from elementary school to MS
- Coding Geek with numerous international awards
- National Badminton Player
- Regular international hackathon participant as team member, team lead and judge
- Passion for volunteer teaching (200+ hours as a volunteer to underprivileged students)

I am confident in my ability to make a valuable research contribution within your group and collaborate actively as a team member. I am eager to explore graduate research opportunities. You may find attached my resume and certificates.

Looking forward to your kind response.

Sincerely,
Ms. Muskaan Irfan
https://www.linkedin.com/in/muskaan03/



Curriculum Vitae

Muskaan Irfan

Data Scientist | Mathematician | Coding Geek | National Badminton Player
Int'l Hackathon Judge | Merit Scholarship Holder
muskaanirfn@gmail.com | LinkedIn | LeetCode | GitHub

PROFILE

- Winner of international competitions, including Stanford, Harvard, and MIT.
- Equally adept in the classroom and on the badminton court.
- Merit scholarship holder since elementary school till MS.
- National Badminton player and international hackathon enthusiast.
- Well-versed in taking failures as lessons and keeping crawling till it is done.
- Comfortable working in cross-cultural, cross-time zone teams, under tight deadlines.
- International Hackathon participant, team lead, judge, and mentor.
- If someone can dream it, I can code it.

ACHIEVEMENTS

- Winner of CS50x Puzzle Day 2024, 2025 & 2026, organized by **Harvard University** (Link)
- Featured by **Tannan FM Magazine** (Fukui, Japan) for contributions to international research and collaborative innovation (Link)
- Selected as **Section Leader/TA** for **Stanford's Code in Place** 2025 (Link).
- Achieved 114th position globally in the **MIT Informatics Tournament 2025,** competing among 530+ teams (Link)
- Ranked **Top 29% globally** out of 15000+ in **Meta Hacker Cup 2025** Round 1 and **6th in Pakistan (Top 3% globally)** in the Practice Round (Link, Link)
- Ranked **2nd** among female coders (overall on 12th rank) in Pakistan in the **Advent of Code 2025** leaderboard (Link)
- Ranked **8th** among female coders in Pakistan in the **Advent of Code 2024** leaderboard (Link)
- Achieved 102nd global rank out of 860+ teams in the **UC Berkeley Coding Competition 2024** (Link)
- Ranked 69th out of 15,000+ Practice Round and 107th out of 22,000+ in Round One in the **Meta Hacker Cup 2024** in Pakistan (Link, Link)
- Won Merit Scholarship from iCodeGuru for six months "AI and Data Analysis Bootcamp" (Link)
- Won Merit Scholarship during Bachelor's degree, awarded by **QAU Merit Scholarship Committee**
- Runner-up in the International Youth Math Challenge
- Achieved Appreciation Award 2022 from Fixit (An NGO dedicated to civic welfare and education) for voluntary contributions to their student mentorship program

EDUCATION

MS Data Science Sep'23 - Sep'25
Air University Pakistan
CGPA: 3.68/4
Relevant Coursework: Machine Learning, BigData Analytics, Data Visualization, Deep Learning, Statistical and Mathematical Methods for Data Analysis, Computer Vision, NLP.

Thesis: *Advancing Human Activity Recognition Through Deep Learning and Sensor Fusion* (PDF)
- Developed an LSTM-based deep learning model for multivariate sensor time-series classification.
- Achieved 97% testing accuracy through advanced preprocessing, feature extraction, and sensor-fusion strategies.
- Designed an open HAR dataset and implemented robust evaluation metrics to validate model performance.

- Expected Paper Submission date: Jan 2026

Awards:
- Won merit scholarship
- Secured 1st position in both *Statistical & Mathematical Methods for Data Analysis* and *Machine Learning* across five Creative Technologies departments.

BS Mathematics Oct'18 - Jun'22
Quaid-i-Azam University, Pakistan
CGPA: 3.2/4
Awards: Won a merit Scholarship.

PUBLICATIONS

1. Sulthan, N., Daryanani, A., Dharmi, S., May, G., **Muskaan**, Sreekumar, G., Saito, R., Amer, S., Ahmed, I., & Aslam, A. (2025). "Pneumonia and COPD in Middle Eastern Older Adults: Current Burden and Future Opportunities", **Published** in *International Journal of Applied Technology in Medical Sciences*, 4(2), 4-25. https://doi.org/10.54878/44jjb542

2. A. Khanzada, **Muskaan,** L. MacCabe, "From Volunteer to Vocation: The Career Impact of Skill and Network Development in a Global Health Technology Nonprofit", **Submitted** to *Journal of International Training and Development* in April 2026. https://doi.org/10.5281/zenodo.19883684

3. A. Khanzada, L. Atlas, N. Rasmussen, F. Khan, **Muskaan,** "Acoustic Phenotyping of Coughs Predicts COVID-19 Screening Accuracy", **Submitted** to *Journal of Artificial Intelligence Research* in August 2026 (PDF*)*

SKILLS

Programming Languages: Python, C++, R
Software & Tools: MATLAB, VS Code, Google Colab, Jupyter Notebook, Spider, GitHub, Tableau, Power BI, WEKA, Overleaf, LaTeX
Framework/ Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, PyTorch, SciPy, Keras, NLTK, Transformers(Hugging Face)
Databases & BigData Technologies: PostgreSQL, PostGIS, PySpark, Apache Hadoop, Apache Hive
Operating Systems: Windows

WORK EXPERIENCE

Mathematics Lecturer Feb'23 - Aug'23
IBA Public School & College Sukkur, Pakistan

- Taught Mathematics, developed curriculum and lesson plans.
- Attended Maths conferences, took advanced Maths courses, and participated in workshops.

Research Intern Aug'22 - Sep'22
National Centre Of Physics (NCP), Pakistan

- Conducted research on **Special Relativity and Cosmology**, using **MATLAB** and **Python** for simulations and modeling, gaining insights into relativistic phenomena.
- Analyzed cosmic microwave background data with CAMB and CLASS to study early universe dynamics, and utilized the Einstein Toolkit for numerical relativity simulations.

VOLUNTEER EXPERIENCE

ML/AI Research Intern June'25 - Present
Virufy (Link) Remote - California, USA

- Conducted independent literature reviews, data analysis, and experimental design to support ongoing research projects, contributing to novel findings and improving the project's scientific direction.
- 2 research papers have been submitted.

Programming/AI Instructor Feb'24 - Present
iCode Guru (Link) Remote - San Jose, CA, USA

- Organized and moderated weekly tech webinars featuring Silicon Valley, California, professional engineers.
- Taught Python, Data Structures and Algorithms, and Maths courses to underprivileged students in their local language.
- Presented LeetCode problems and responded to programming queries. Guided students to participate in International Hackathons and Coding Competitions.

Section Leader/ Teaching Assistant (Link) Apr'25 - May'25
Stanford University Remote - Palo Alto, CA, USA

- Taught international students Python using Karel the Robot at Stanford.
- Helped peer mentors discuss new teaching methodologies.

Tech Mentor & Judge (Link) Aug'24 - Present
GenAI Hackathon Pak-Angels, San Jose CA, USA

- Delivered a webinar to 400+ participants, covering hackathon strategies, project ideation, teamwork, and effective pitching techniques.
- Guided teams in developing Gen-AI applications, including RAG models and chatbots, ensuring smooth project execution and deployment.
- Evaluated 20+ projects and conducted one-on-one mentorship sessions, addressing technical queries, refining ideas, and ensuring fair recognition of outstanding work.

INTERNATIONAL HACKATHONS

Cohere Hackathon - LabLab.ai (Presentation) (Link) Jan'23
Team Size: 4 | Muskaan (Team Lead, Pakistan)**,** Leandra P.(USA), Maha Z.(Pakistan), Ibtisam M.(India)

My Role: Project Manager & Idea Lead
Tech Stacks: Python, Cohere API (LLMs), OpenCV, NLP, Computer Vision, Sentiment Analysis

- Developed an AI Job Interview Assistant using the Cohere API, enabling candidates to record responses and analyze their verbal answers and facial expressions.
- Enhanced candidate experience and employer brand by building confidence, saving time, and streamlining the interview process.

OpenAI Hackathon - LabLab.ai (Presentation) (Link) Dec'22
Team Size: 6 | Muskaan (Pakistan), Nargiz S.(Team Lead, Georgia), Nik Chein(Taiwan), Ayesha A.(UAE), Hafsa N.(Pakistan), Nafia A.(Pakistan)

My Role: AI Product Contributor & Developer

Tech Stack: Node.js, Python, OpenAI API (LLMs), Google Books API, Recommender Systems

- Developed the Bibliotopia App, an AI-powered bookshop leveraging GPT-3 to provide personalized book recommendations and promote literacy initiatives.
- Collaborated with humanitarian organizations, Google Books, government agencies, and libraries to drive literacy programs, managing product development, AI engineering, and customer service.

CONFERENCES, WORKSHOPS, AND WEBINARS

- Conducted 6-hour workshops on *Python Data Visualization for Mathematicians*, focusing on Matplotlib, Seaborn, and practical charting techniques at iCodeGuru (Link)
- Delivered a webinar, *Beginner's Guide to Join Hackathons* at iCodeGuru & Aspire-Pak (Link)
- Organized a Session on *Unity in Diversity and Pluralism by Centre for Peace Education Pakistan* (Link)
- Organized workshop on *Startup for Applied Researchers in Pakistan* (Link)

HOBBIES/INTERESTS

- **Hackathons:** Actively participate in international hackathons, collaborating with diverse teams to create innovative projects.
- **Writing:** Share insights on technology, data science, and algorithms through articles on LinkedIn, engaging a wide audience.
- **Badminton:** Competed at the National Olympics, proudly representing Quaid-i-Azam University, demonstrating high-level athletic performance, discipline, and team spirit.
- **Artistic Skills:** Artistic pursuits include painting, calligraphy, portrait sketching, and creating custom artwork. (Link)

Internships
Int'l Hackathons
Int'l Competitions

Featured in Japan's Tannan FM Magazine

Honored to be featured in Japan's Tannan FM Magazine for contributions to a research paper and community impact.



Research Internship

(Aug 2022 - Sep 2022)

- Only 2 students were selected from Physics and Applied Mathematics departments.
- I was selected because I secured **first position in the Special Relativity course**.
- During the internship:
    - Contributed to simulations and modelling in MATLAB and Python.
    - Gained hands-on research experience in cosmology.

Certificate from the National Centre for Physics, Islamabad: available on request.

Harvard CS50X Puzzle Day 2025

- Won first position by solving all 9 puzzles.



CS50 Certificate

CS50 congratulates

Muskaan

whose team solved 9 out of 9 puzzles in CS50x Puzzle Day 2025.

Awarded from Cambridge, Massachusetts,
in the year two thousand twenty-five.

David J. Malan
Gordon McKay Professor of the Practice of Computer Science
Harvard University

https://cs50.harvard.edu/certificates/fe09e643-08a1-4577-ba9b-980a8abc235b

Harvard CS50X Puzzle Day 2024

- Won first position by solving all 9 puzzles.



CS50 Certificate

CS50 congratulates

Muskaan

whose team solved 9 out of 9 puzzles in CS50x Puzzle Day 2024.

Awarded from Cambridge, Massachusetts,
in the year two thousand twenty-four.

David J. Malan
Gordon McKay Professor of the Practice of Computer Science
Harvard University

https://cs50.harvard.edu/certificates/a61cf071-87ed-41a2-a153-d4f76ba8de2c

Meta Hacker Cup 2025

- 6th in Pakistan (Top 3% globally) in the Practice Round.



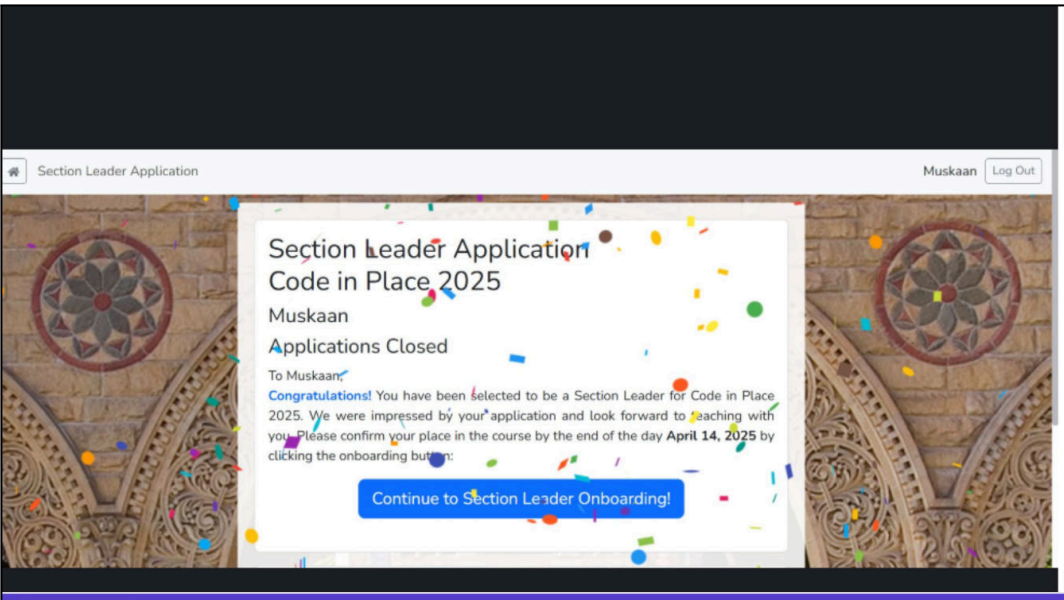
Meta Hacker Cup 2024

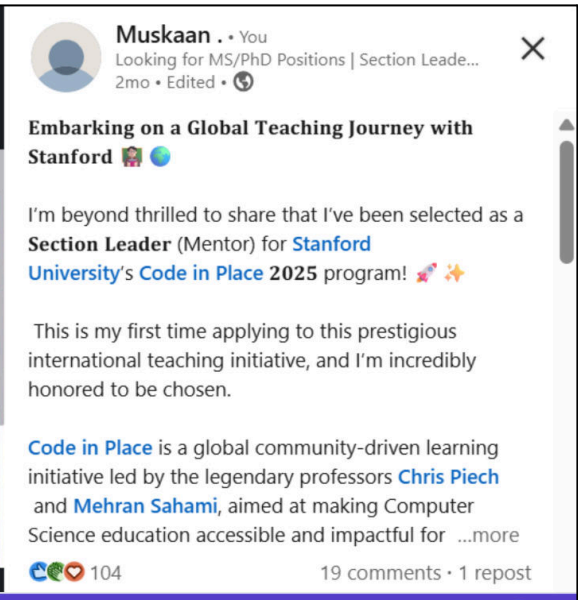
- **Top 84%** of Round 1 participants.

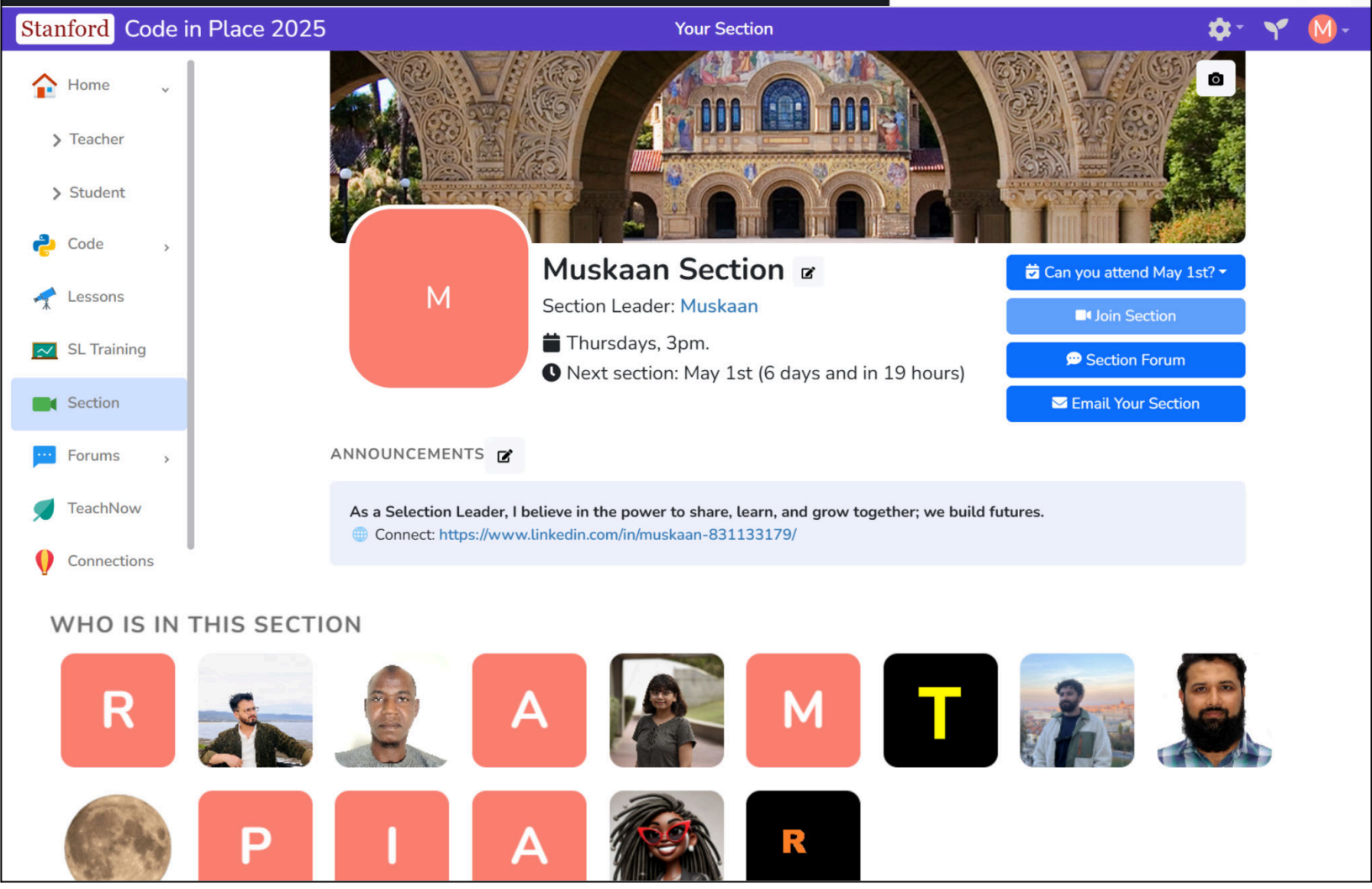


Section Leader @ Stanford Code In Place (CS 106A)

- Taught Python to international students (online)







Teaching Experience: Python for Mathematicians

(Oct 2024)

- Data Preprocessing using libraries
- Machine Learning Algorithms
- Linear Algebra and Its Applications
- Data Structure and Algorithms
- Data Visualization



Judge & Mentor of GenAI Int'l Hackathon

April - May 2025, San Jose, CA, USA

- Helped students build RAG models and chatbots
- Reviewed 20+ projects and gave 1:1 support to improve ideas and ensure fair judging



Judge & Mentor of GenAI Int'l Hackathon

Dec - Jan 2025, San Jose, CA, USA



UC Berkeley: Calico Coding Competition 2024

(Aug 2024)

- Achieved 102nd global rank out of 860+ teams.





RANK	TEAM	SCORE	1	2	2a1	3	4	5	5a1	6	6a1	7	7
98	IANA	30 366	17 2 tries	23 1 try	32 1 try	37 1 try	26 1 try	90 5 tries	2 tries	6 tries		91 1 try	
99	Kalam Tech	30 380	21 1 try	31 1 try	119 1 try	36 1 try	48 1 try	46 1 try	2 tries	2 tries	2 tries	79 1 try	0 + 1
100	WarriorsX	30 387	37 2 tries	46 1 try	78 1 try	44 1 try	51 1 try	49 1 try	3 + 1 tries	3 tries	1 + 1 tries	72 1 try	
101	CodeX	30 390	20 1 try	45 1 try	115 1 try	43 1 try	47 1 try	48 1 try	0 + 1 tries	0 + 1 tries		72 1 try	
102	Code Crushers	30 419	25 1 try	82 2 tries	119 1 try	23 1 try	42 1 try	47 1 try	4 + 4 tries	10 tries		71 1 try	
103	Neural Ninjas	30 446	39 1 try	39 4 tries	79 3 tries	54 2 tries	51 1 try	57 1 try	4 tries	7 tries	0 + 2 tries	67 1 try	1

OpenAI International Hackathon

(Dec 2022)

Team Size: 6 | Muskaan (Pakistan), Nargiz S.(Team Lead, Georgia), Nik Chein(Taiwan), Ayesha A.(UAE), Hafsa N.(Pakistan), Nafia A.(Pakistan)

My Role: AI Product Contributor & Developer
Tech Stack: Node.js, OpenAI API (LLMs), Google Books API, Recommender Systems, Python

- Built *Bibliotopia*, an AI-powered bookshop.
- Partnered with NGOs and agencies to support literacy programs.
- In addition to providing technical support to junior team members, I personally worked on AI pipeline, integrated key APIs, and built the core recommendation system.



Cohere International Hackathon

(Jan 2023)

Team Size: 4 | Muskaan (Team Lead, Pakistan)**,** Leandra P.(USA), Maha Z.(Pakistan), Ibtisam M.(India)

My Role: Project Manager & Idea Lead
Tech Stacks: Python, Cohere API (LLMs), OpenCV, NLP, Computer Vision, Sentiment Analysis

- In addition to guiding the whole team, I myself contributed towards architecting the system's AI engine and leading the development of its intelligent feedback features.
- Developed an AI Job Interview Assistant using **Cohere API** to analyze verbal responses and facial expressions.
- Streamlined interviews by delivering automated real-time feedback.



Runner-Up: International Youth Math Challenge

- Represented my University during my BS in this competition
- Won prize money for my performance.



QR-2021-EDD8716E582

Date: 25. October 2021

International Youth Math Challenge

Participation Certificate

presented to

Muskaan

Pakistan

for a successful participation in the qualification round of 2021. The International Youth Math Challenge is an independent international mathematics competition for high school and university students with the aim to encourage young students to engage in a variety of mathematical areas.

Fabian Schneider, International Coordinator

Rami Aly, Public Relations Manager

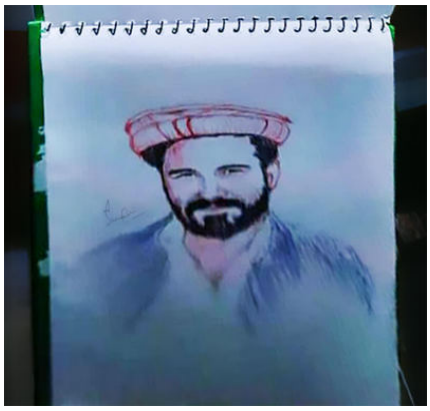
Volunteer Work: Certificate of Appreciation

- Awarded the **Fixit Appreciation Award** for teaching students in community programs.
- Helped students improve their academic basics and gain confidence in their studies.



Calligraphy, Portrait & Sketches

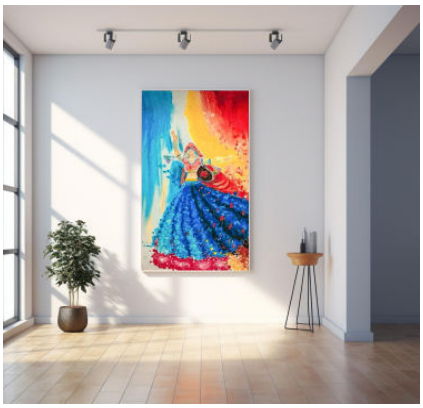
I loved making sketches from a young age, and during my BS, I used them to self-fund my education. I worked with organizations to make on-demand portraits, calligraphy, and sketches: artist-portal



Sketch of Gratitude, 2022



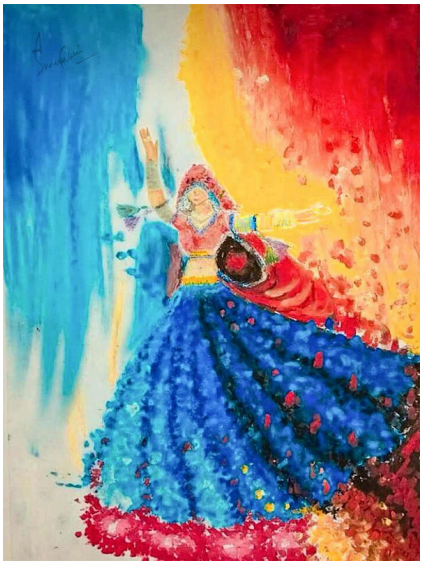
Captured in Pencil, 2021



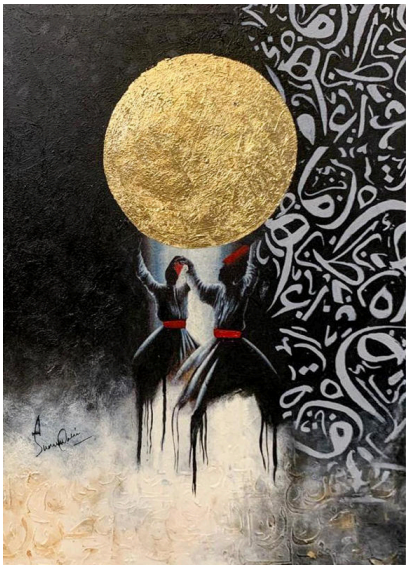
Spectrum Woman, 2023



Dr. Zafar's Portrait, 2025



Story in Hues, 2023



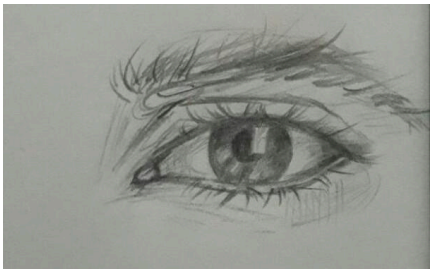
Echoes of Mystic Night, 2025



Honoring the Teacher, 2023



Letters with Soul, 2024



Moment Frozen in Graphite, 2025

A Girl with Big Dreams

"Excellence is not an act, but a habit."

My journey is defined by a balance of academic rigour and competitive spirit. As a Merit Scholarship holder since elementary school, I learned early on that consistency is the key. This foundation allowed me to step into leadership roles, serving as **Head Girl/Prefect** and actively hosting large-scale university events and technical presentations.



But my discipline is truly tested on the court. Since college, I have played badminton at **state-level tournaments,** competing in **21 matches** and **winning 19**. As a National Badminton Player, I have represented my university at the **National Olympics and state-level tournaments.** Whether organizing a team event or competing for a match point, I bring the same level of focus, strategy, and resilience that I apply to my code.











"I aim not just to learn, but to create, challenge, and contribute to the frontiers of knowledge."